Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Multi-arm, Open-label Phase I/IIa Study to Assess the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics and Preliminary Efficacy of AZD5305 in Combination with New Hormonal Agents in Patients with Metastatic Prostate Cancer **Short Title/Acronym:** PETRANHA **Protocol Number:** D9720C00003 **NCT Number:** NCT05367440 **Posting ID:** 105390778972234523 **Sponsor/Company Name:** AstraZeneca **Trial Status:** Recruiting **Investigational Product:** AZD5305 (saruparib) in combination with New Hormonal Agents (Enzalutamide, Abiraterone Acetate, Darolutamide, or Apalutamide) **Phase of Development:** Phase I/IIa **Medical Monitor:** AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, pharmacokinetics (PK), pharmacodynamics (PD), and preliminary efficacy of AZD5305 when given in combination with new hormonal agents (NHAs) in patients with Metastatic Prostate Cancer. **Secondary Objectives:** To assess the overall PK parameters (AUC, Cmax, tmax) of AZD5305, as well as pharmacodynamics and preliminary efficacy of the combination therapy.

2.2 Background & Rationale To address the need for improved and more tolerable treatments in advanced prostate cancer. PARP inhibitors combined with androgen receptor pathway inhibitors (NHAs) improve outcomes, but current non-selective PARP inhibitors often present toxicity challenges. AZD5305 (saruparib) is a new-generation, potential best-in-class PARP inhibitor that selectively targets and traps PARP1 while having minimal effect on PARP2. This selectivity may offer a significantly improved therapeutic window, allowing for safer combination with NHAs to halt cancer cell repair and slow tumor growth in both metastatic castration-resistant (mCRPC) and castration-sensitive prostate cancer (mCSPC) patients.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm per cohort (Multi-arm dose escalation and expansion) **Blinding:** Open-label **Randomization:** N/A (Non-randomized; patients are assigned to 1 of 4 treatment arms based on the specific NHA chosen by the investigator) **Study Structure:** The study consists of 2 parts: - **Part A (Dose Escalation):** Includes patients with mCRPC or mCSPC across 4 individual arms evaluating safety, tolerability, and preliminary efficacy. - **Part B (Dose Expansion):** Includes patients with mCSPC only across up to 4 individual arms to build on safety and preliminary efficacy data.

3.2 Planned Enrollment & Duration Target N: 175 targeted enrollment (Approximately 783 patients will be screened globally to ensure the required number of evaluable patients; up to 356 screened for 308 evaluable in Part A, and up to 427 screened for 360 evaluable in Part B). **Number of Sites:** Global multicenter trial **Estimated Study Start:** [DD-MMM-YYYY] **Estimated Primary Completion:** [DD-MMM-YYYY]

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male patients ≥ 18 years old at the time of screening. 2. Histologically confirmed diagnosis of metastatic prostate cancer and a candidate for treatment with enzalutamide, abiraterone acetate, darolutamide, or apalutamide. 3. Surgically or medically castrated. 4. Adequate organ and marrow function. 5. Eastern Cooperative Oncology Group Performance Status (ECOG PS) of 0-1 with no deterioration over the previous 2 weeks. 6. Life expectancy ≥ 16 weeks. 7. Non-sterilized male patients sexually active with a female partner of childbearing potential must use a condom with spermicide from screening to approximately 6 months post-treatment. 8. **Pharmacodynamic (PD) Cohort specific:** Must have at least 1 tumor suitable for paired biopsies. 9. **Part A specific:** Patients with mCRPC or mCSPC. 10. **Part B specific:** Patients must have mCSPC (de novo or recurrent) with a baseline PSA value of ≥ 0.2 ng/mL.

4.2 Exclusion Criteria 1. **Prior Therapies:** - *For Part A mCRPC:* Any previous treatment with an NHA (prior NHA also excluded for PD cohorts), PARP inhibitor, Lutetium-PSMA, or platinum chemotherapy. - *For Part A and B mCSPC:* Any previous treatment with a PARPi, platinum, NHA, Immuno-oncology (IO), radiopharmaceutical therapy, or prior docetaxel in the mCSPC setting. 2. **Concomitant Medications:** Strong and moderate CYP3A4 inducers/inhibitors (all arms); strong CYP2C8 inhibitors (Arm 1); strong P-glycoprotein inducers (Arm 3). Drugs known to prolong/ shorten QT and have a risk of Torsades de Pointes. 3. **Recent Treatments:** Investigational agents within 5 half-lives or 3 weeks; cytotoxic/non-cytotoxic treatments within 3 weeks or 5 half-lives; biological/IO products within 4 weeks; live virus/ bacterial vaccines within 28 days; major surgery within 4 weeks; wide-field radiotherapy within 4 weeks (or limited field for palliation within 2 weeks). 4. **Unresolved Toxicities:** > CTCAE Grade 1 from prior therapy (except alopecia and peripheral neuropathy). Any history of severe pancytopenia lasting > 2 weeks. 5. **Disease & Co-morbidities:** - History of myelodysplastic syndrome (MDS) or acute myeloid leukaemia (AML). - Spinal cord compression or brain metastases (unless asymptomatic, treated, and stable). - Severe or uncontrolled systemic diseases, active bleeding diatheses, or active infections (including HIV, Hepatitis B, and C). - Predisposition to bleeding (e.g., active peptic ulcer, hemorrhagic stroke within 6 months, proliferative diabetic retinopathy). - Clinically significant cardiovascular/ cardiac diseases (e.g., symptomatic heart failure, uncontrolled hypertension, acute coronary syndrome, QT prolongation, abnormal ECG, cardiomyopathy, valvular heart disease, atrial fibrillation, stroke). - Refractory nausea/vomiting, chronic GI diseases, inability to swallow, or significant bowel resection. Active interstitial lung disease (ILD). - History of another primary malignancy within 3 years (except for malignancies treated with curative intent and low recurrence risk). 6. **Arm-Specific Exclusions:** - *Arm 1 (Enzalutamide) & Arm 4 (Apalutamide):* History of seizure or predisposing conditions (e.g., cortical stroke, brain trauma). - *Arm 2 (Abiraterone acetate):* Active infection/ condition contraindicating systemic steroids, low serum potassium (< 3.5 mmol/L), or history of uncontrolled pituitary/adrenal dysfunction. - *Arm 4 (Apalutamide):* Moderate or severe skin conditions (e.g., scleroderma, lupus) or conditions increasing skin toxicity risk.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Daily oral administration of AZD5305 evaluated in combination with an NHA (Arm 1: Enzalutamide; Arm 2: Abiraterone Acetate; Arm 3: Darolutamide; Arm 4: Apalutamide). **Route of Administration:** Oral **Duration of Treatment:** Continuous daily treatment until disease progression, initiation of alternative anticancer therapy, unacceptable toxicity, withdrawal of consent, or other discontinuation reasons occur.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for prostate cancer; systemic steroids where required (e.g., prednisone alongside Abiraterone Acetate). **Prohibited:** Concurrent investigational therapies, Lu-PSMA, alternate platinum chemotherapies, strong/moderate CYP3A4 inhibitors/inducers, and drugs with QT/Torsades de Pointes risks.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, ECG, Labs, Baseline Imaging
Baseline	Day 0	Enrollment, First Dose, Safety Baseline Check
Interim	Cycle 1 to X	Safety Labs, PK/PD blood sampling, Tumor Assessment Scans (Biopsies for PD cohort)
End of Study	EOT	Final Vital Signs, Efficacy Assessment, IP Accountability

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Adverse Events:** Number of patients with Adverse Events (AEs) and Serious Adverse Events (SAEs), including abnormal clinical observations, abnormal ECG parameters, abnormal laboratory assessments, and abnormal vital signs changed from baseline. 2. **Dose-Limiting Toxicities (Part A):** Number of patients with Dose-Limiting Toxicities (DLTs) to assess the safety and tolerability of AZD5305 when given in combination with NHAs.

7.2 Secondary Endpoints 1. Area Under the concentration Curve (AUC) of AZD5305. 2. Maximum plasma concentration (Cmax) of AZD5305. 3. Time to maximum concentration (tmax) of AZD5305.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient visits, safety labs, and ECGs. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Safety Review Committee utilizing emerging safety and PK data to guide dose escalation and expansion cohorts.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving ≥ 1 dose of study drug); Efficacy Evaluable Set. **Sample Size Justification:** Up to ~783 screened to enroll ~175 target patients (yielding up to 308 evaluable for Part A and 360 evaluable for Part B), powered appropriately for early-phase dose-escalation and cohort-expansion safety thresholding. **Handling of Missing Data:** Missing values evaluated as observed for primary safety endpoints without imputation.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits ensuring data integrity and adherence to protocol. **Audit Trail:** Data tracked via eCRFs with rigorous query generation and resolution standards.

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11. Study Identifiers & Governance

Primary ID: D9720C00003 **Registry Number:** NCT05367440 **Posting ID:** 105390778972234523 **Sponsor/Lead Organization:** AstraZeneca **Study

Title: PETRANHA (Study of AZD5305 When Given in Combination With New Hormonal Agents in Patients With Metastatic Prostate Cancer) **Official Regulatory Title:** A Multi-arm, Open-label Phase I/IIa Study to Assess the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics and Preliminary Efficacy of AZD5305 in Combination with New Hormonal Agents in Patients with Metastatic Prostate Cancer

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12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: Metastatic Prostate Cancer (mCRPC and mCSPC) | *UMLS Preferred Name: Prostatic Neoplasms* **Diagnosis Threshold:** Histologically confirmed metastatic disease **Medical Methodology:** Targeted Therapy (Selective PARP1 Inhibitor) + Guideline Determined New Hormonal Agents **Optimization Target:** Safe combination dosage (MTD/RP2D)

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13. Study Procedures & Methodology

Management Pathway: Standard oncology sequence for advanced metastatic prostate cancer **Education Protocol (HCP):** Investigator training on selective PARP inhibition and AE management **Education Protocol (Patient):** Informed consent, daily oral dosing adherence, and reporting of GI/cardiac symptoms, safe sexual practices (condom & spermicide) **Quality Audit Mechanism:** Central imaging review for tumor assessment & real-time PK/PD safety data monitoring

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14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, initiation of alternative therapy, or withdrawal of consent. **Onsite Visit Cadence:** Every study cycle (e.g., Q4W) for dispensing and safety labs **Remote Monitoring Frequency:** As indicated per site protocol **Checklist Review Interval:** Every cycle for concurrent medication and pill diary review

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]